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Time to Practice: Hierarchical Clustering
Graded Assignment • 15 min

Due Oct 19, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. A data scientist is using hierarchical clustering to organize documents based on similarity and creates a dendrogram. If the goal is to have a detailed categorization, at what approximate level should the tree be cut? 1 / 1 point

- (•) Lower level
- ( ) Upper level
- ( ) At the root

Nice work

Cutting the dendrogram at a lower level results in more clusters, providing a more detailed and fine-grained categorization of the documents.

2. When considering whether to merge two clusters, what does calculation does Ward's method primarily involve? 1 / 1 point

- (•) Sum of squared distances
- ( ) Number of points in clusters
- ( ) Maximum distance between clusters
- ( ) Average linkage distance

Nice work

Ward's method compares the total sum of squared distances to the cluster centroid within each cluster when the clusters are separate versus the same total sum of squared distances when they are merged into one cluster, to determine the cost of merging the clusters.

3. Which of the following statements about the group average criterion for hierarchical clustering is *incorrect*? 1 / 1 point

- (•) The group average criterion favors elongated shapes over more globular clusters.
- ( ) Group average is less susceptible to noise and outliers.
- ( ) Group average clustering tends to favor more globular clusters
- ( ) Group average can be considered a compromise between single linkage and complete linkage criteria.

Nice work

That's right, this fact is not true: The group average method calculates the average distance between two potential clusters, providing a compromise that reduces susceptibility to noise and outliers while still favoring more globular clusters over elongated shapes.

4. Which of the following statements about hierarchical clustering methods is *incorrect*? 1 / 1 point

- ( ) Top-down clustering can be sensitive to early errors: a bad first split choice can ruin the entire process.
- ( ) Bottom-up clustering doesn't make use of the entire dataset and so may miss information that would lead to better clustering decisions overall.
- (•) Ward's method tends to lead to unequal-sized clusters or even single-point outlier clusters if noise is present.
- ( ) The single linkage criterion computes the minimum distance between two clusters.

Nice work

That's right - this claim is false: Ward's method actually tends to give more equal-sized clusters and is quite robust to noise.

5. Which of the following is a characteristic key operation in a bottom-up agglomerative clustering process? 1 / 1 point

- (•) Computation of the proximity matrix
- ( ) Assignment of data points to clusters
- ( ) Initialization of cluster centers

Nice work

Correct: The proximity matrix, which tracks the distances between clusters, is crucial for determining which clusters to merge at each step in bottom-up agglomerative clustering.

6. How can the number of clusters appropriate for a dataset be determined from a dendrogram? 1 / 1 point

- (•) By analyzing the height at which clusters merge
- ( ) By measuring the maximum width of the dendrogram
- ( ) By counting the number of leaves in the dendrogram

Nice work

The number of clusters can be determined by analyzing the height at which clusters merge in a dendrogram, with significant heights indicating distinct clusters that were only merged when far apart.